

Data_Overview

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BIOS6624 Project 2 - Script 01: Preliminary Data Overview

Purpose:

- Read preliminary dataset
- Examine structure and summary statistics
- Review plots of variables
- Calculate skewness
- Estimate correlations among predictors and outcomes

Outputs:

Results/Tables/summary_table.csv

Results/Tables/skewness_table.csv

Results/Tables/correlation_table.csv

Results/Figures/*.png

```
# -----  
# Clean environment  
# -----  
rm(list = ls())  
  
# -----  
# Load libraries  
# -----  
library(tidyverse)
```

```
## Warning: package 'tidyverse' was built under R version 4.5.2
```

```
## Warning: package 'ggplot2' was built under R version 4.5.3
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v dplyr      1.1.4      v readr      2.1.5
```

```
## v forcats    1.0.0      v stringr   1.5.1
```

```
## v ggplot2     4.0.2      v tibble    3.3.0
```

```
## v lubridate  1.9.4      v tidyr     1.3.1
```

```
## v purrr      1.1.0
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(readr)
```

```
library(e1071)
```

```
##
```

```
## Attaching package: 'e1071'
```

```
##
```

```
## The following object is masked from 'package:ggplot2':
```

```
##
```

```
##      element
```

```
# -----
```

```
# Define directories
```

```
# -----
```

```
prelim_path <- "C:/Users/mojde/Desktop/Documents/BIOS-6624-Projects/Project2/DataPrelim/PrelimData.csv"
```

```
results_tbl <- "C:/Users/mojde/Desktop/Documents/BIOS-6624-Projects/Project2/Results/Tables"
```

```
results_fig <- "C:/Users/mojde/Desktop/Documents/BIOS-6624-Projects/Project2/Results/Figures"
```

```
output_dir <- "C:/Users/mojde/Desktop/Documents/BIOS-6624-Projects/Project2/Output"
```

```
# Create folders if they don't exist
```

```
dir.create(results_tbl, recursive = TRUE, showWarnings = FALSE)
```

```
dir.create(results_fig, recursive = TRUE, showWarnings = FALSE)
```

```
dir.create(output_dir, recursive = TRUE, showWarnings = FALSE)
```

```
# -----
```

```
# Read raw data, blank as NA
```

```
# -----
```

```
data <- read_csv(prelim_path, na = c("", " ", "NA"))
```

```
## Rows: 30 Columns: 4
```

```
## -- Column specification -----
```

```
## Delimiter: ","
```

```
## dbl (4): CVLT_CNG3, CORT_CNG3, IL_6, MCP_1
```

```
##
```

```
## i Use 'spec()' to retrieve the full column specification for this data.
```

```
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
# -----
# Inspect structure
# -----
str(data)
```

```
## spc_tbl_ [30 x 4] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
## $ CVLT_CNG3: num [1:30] -0.48 -0.56 0.22 -1.29 -1.46 -3.17 -1.33 -2.24 -2.33 -1.6 ...
## $ CORT_CNG3: num [1:30] -1.13 -7.19 -4.82 -3.33 -4.08 -5.56 -5.77 -5.51 -5.01 -6.87 ...
## $ IL_6      : num [1:30] 1.66 7.91 7 5.31 6.31 ...
## $ MCP_1     : num [1:30] 7.27 12.97 9.58 9.01 10.26 ...
## - attr(*, "spec")=
## .. cols(
## ..   CVLT_CNG3 = col_double(),
## ..   CORT_CNG3 = col_double(),
## ..   IL_6 = col_double(),
## ..   MCP_1 = col_double()
## .. )
## - attr(*, "problems")=<externalptr>
```

```
summary(data)
```

	CVLT_CNG3	CORT_CNG3	IL_6	MCP_1
## Min.	:-3.860	Min. :-7.290	Min. : 0.760	Min. : 3.700
## 1st Qu.:	-2.152	1st Qu.: -5.758	1st Qu.: 4.590	1st Qu.: 9.197
## Median :	-1.290	Median :-5.190	Median : 6.395	Median :10.965
## Mean :	-1.400	Mean :-4.960	Mean : 6.503	Mean :10.881
## 3rd Qu.:	-0.625	3rd Qu.: -4.242	3rd Qu.: 8.920	3rd Qu.:13.225
## Max. :	0.970	Max. :-1.130	Max. :11.300	Max. :15.500

```
# -----
# Create log variables
# -----
data$log_IL6 <- log(data$IL_6 + 0.001)
data$log_MCP1 <- log(data$MCP_1 + 0.001)

# -----
# Summary table
# -----
summary_table <- data.frame(
  variable = c("IL_6", "MCP_1", "CVLT_CNG3", "CORT_CNG3"),
  n = c(sum(!is.na(data$IL_6)),
        sum(!is.na(data$MCP_1)),
        sum(!is.na(data$CVLT_CNG3)),
        sum(!is.na(data$CORT_CNG3))),
  mean = c(mean(data$IL_6, na.rm = TRUE),
            mean(data$MCP_1, na.rm = TRUE),
            mean(data$CVLT_CNG3, na.rm = TRUE),
            mean(data$CORT_CNG3, na.rm = TRUE)),
  sd = c(sd(data$IL_6, na.rm = TRUE),
          sd(data$MCP_1, na.rm = TRUE),
          sd(data$CVLT_CNG3, na.rm = TRUE),
```

```

        sd(data$CORT_CNG3, na.rm = TRUE))
)

print(summary_table)

```

```

##      variable  n      mean      sd
## 1      IL_6 30  6.502667 2.792569
## 2      MCP_1 30 10.881000 2.813424
## 3 CVLT_CNG3 30 -1.399667 1.078810
## 4 CORT_CNG3 30 -4.960000 1.417134

```

```

write.csv(summary_table, file.path(results_tbl, "summary_table.csv"), row.names = FALSE)

```

```

# -----
# Skewness table
# -----
skewness_table <- data.frame(
  variable = c("IL_6", "MCP_1", "log_IL6", "log_MCP1", "CVLT_CNG3", "CORT_CNG3"),
  skewness = c(
    skewness(data$IL_6, na.rm = TRUE),
    skewness(data$MCP_1, na.rm = TRUE),
    skewness(data$log_IL6, na.rm = TRUE),
    skewness(data$log_MCP1, na.rm = TRUE),
    skewness(data$CVLT_CNG3, na.rm = TRUE),
    skewness(data$CORT_CNG3, na.rm = TRUE)
  )
)

print(skewness_table)

```

```

##      variable  skewness
## 1      IL_6 -0.05197042
## 2      MCP_1 -0.30461350
## 3    log_IL6 -1.52014680
## 4    log_MCP1 -1.28327773
## 5 CVLT_CNG3 -0.32159566
## 6 CORT_CNG3  0.66082727

```

```

write.csv(skewness_table, file.path(results_tbl, "skewness_table.csv"), row.names = FALSE)

```

```

# -----
# Combined histograms
# -----

png(file.path(results_fig, "all_histograms.png"), width = 1600, height = 1200, res = 150)

par(mfrow = c(2, 3), mar = c(4, 4, 3, 1), oma = c(0, 0, 2, 0))

hist(data$IL_6,
      main = "IL-6",
      xlab = "IL-6",

```

```

    col = "gray80",
    border = "white")

hist(data$log_IL6,
     main = "log(IL-6)",
     xlab = "log(IL-6)",
     col = "gray80",
     border = "white")

hist(data$MCP_1,
     main = "MCP-1",
     xlab = "MCP-1",
     col = "gray80",
     border = "white")

hist(data$log_MCP1,
     main = "log(MCP-1)",
     xlab = "log(MCP-1)",
     col = "gray80",
     border = "white")

hist(data$CVLT_CNG3,
     main = "CVLT Change",
     xlab = "CVLT_CNG3",
     col = "gray80",
     border = "white")

hist(data$CORT_CNG3,
     main = "Cortical Change",
     xlab = "CORT_CNG3",
     col = "gray80",
     border = "white")

mtext("Histograms of Inflammatory Markers and Outcome Change Scores",
      outer = TRUE, cex = 1.4)

dev.off()

```

```

## pdf
## 2

```

```

# -----
# Combined scatterplots
# -----

png(file.path(results_fig, "all_scatterplots.png"), width = 2000, height = 1600, res = 150)

par(mfrow = c(2, 4), mar = c(4, 4, 3, 1), oma = c(0, 0, 2, 0))

plot(data$IL_6, data$CVLT_CNG3,
     xlab = "IL-6", ylab = "CVLT Change",
     main = "IL-6 vs CVLT",
     pch = 19, col = "gray40")

```

```

abline(lm(CVLT_CNG3 ~ IL_6, data = data), col = "blue", lwd = 2)

plot(data$IL_6, data$CORT_CNG3,
      xlab = "IL-6", ylab = "Cortical Change",
      main = "IL-6 vs Cortical",
      pch = 19, col = "gray40")
abline(lm(CORT_CNG3 ~ IL_6, data = data), col = "blue", lwd = 2)

plot(data$MCP_1, data$CVLT_CNG3,
      xlab = "MCP-1", ylab = "CVLT Change",
      main = "MCP-1 vs CVLT",
      pch = 19, col = "gray40")
abline(lm(CVLT_CNG3 ~ MCP_1, data = data), col = "blue", lwd = 2)

plot(data$MCP_1, data$CORT_CNG3,
      xlab = "MCP-1", ylab = "Cortical Change",
      main = "MCP-1 vs Cortical",
      pch = 19, col = "gray40")
abline(lm(CORT_CNG3 ~ MCP_1, data = data), col = "blue", lwd = 2)

plot(data$log_IL6, data$CVLT_CNG3,
      xlab = "log(IL-6)", ylab = "CVLT Change",
      main = "log(IL-6) vs CVLT",
      pch = 19, col = "gray40")
abline(lm(CVLT_CNG3 ~ log_IL6, data = data), col = "blue", lwd = 2)

plot(data$log_IL6, data$CORT_CNG3,
      xlab = "log(IL-6)", ylab = "Cortical Change",
      main = "log(IL-6) vs Cortical",
      pch = 19, col = "gray40")
abline(lm(CORT_CNG3 ~ log_IL6, data = data), col = "blue", lwd = 2)

plot(data$log_MCP1, data$CVLT_CNG3,
      xlab = "log(MCP-1)", ylab = "CVLT Change",
      main = "log(MCP-1) vs CVLT",
      pch = 19, col = "gray40")
abline(lm(CVLT_CNG3 ~ log_MCP1, data = data), col = "blue", lwd = 2)

plot(data$log_MCP1, data$CORT_CNG3,
      xlab = "log(MCP-1)", ylab = "Cortical Change",
      main = "log(MCP-1) vs Cortical",
      pch = 19, col = "gray40")
abline(lm(CORT_CNG3 ~ log_MCP1, data = data), col = "blue", lwd = 2)

mtext("Scatterplots of Inflammatory Markers and Outcome Change Scores",
      outer = TRUE, cex = 1.4)

dev.off()

```

```

## pdf
## 2

```

```
# Reset plotting layout
par(mfrow = c(1, 1))
```

```
# -----
#Correlation table
# -----
# IL-6 vs CVLT
t1 <- cor.test(data$IL_6, data$CVLT_CNG3, use = "complete.obs")
# IL-6 vs Cortical Thickness
t2 <- cor.test(data$IL_6, data$CORT_CNG3, use = "complete.obs")
# MCP-1 vs CVLT
t3 <- cor.test(data$MCP_1, data$CVLT_CNG3, use = "complete.obs")
# MCP-1 vs Cortical Thickness
t4 <- cor.test(data$MCP_1, data$CORT_CNG3, use = "complete.obs")
# IL-6 vs MCP-1
t5 <- cor.test(data$IL_6, data$MCP_1, use = "complete.obs")
# CVLT vs Cortical Thickness
t6 <- cor.test(data$CVLT_CNG3, data$CORT_CNG3, use = "complete.obs")
correlation_table <- data.frame(
  comparison = c(
    "IL_6 vs CVLT_CNG3",
    "IL_6 vs CORT_CNG3",
    "MCP_1 vs CVLT_CNG3",
    "MCP_1 vs CORT_CNG3",
    "IL_6 vs MCP_1",
    "CVLT_CNG3 vs CORT_CNG3"
  ),
  r = c(t1$estimate, t2$estimate, t3$estimate, t4$estimate, t5$estimate, t6$estimate),
  lower = c(t1$conf.int[1], t2$conf.int[1], t3$conf.int[1], t4$conf.int[1], t5$conf.int[1], t6$conf.int[1]),
  upper = c(t1$conf.int[2], t2$conf.int[2], t3$conf.int[2], t4$conf.int[2], t5$conf.int[2], t6$conf.int[2])
)

print(correlation_table)
```

##	comparison	r	lower	upper
## 1	IL_6 vs CVLT_CNG3	-0.2585617	-0.5660978	0.11215500
## 2	IL_6 vs CORT_CNG3	-0.5993366	-0.7891998	-0.30490277
## 3	MCP_1 vs CVLT_CNG3	-0.3183585	-0.6088014	0.04734037
## 4	MCP_1 vs CORT_CNG3	-0.6852730	-0.8385249	-0.43154477
## 5	IL_6 vs MCP_1	0.9335558	0.8638145	0.96819171
## 6	CVLT_CNG3 vs CORT_CNG3	0.1559104	-0.2165208	0.48872788

```
write.csv(correlation_table, file.path(results_tbl, "correlation_table.csv"), row.names = FALSE)
```